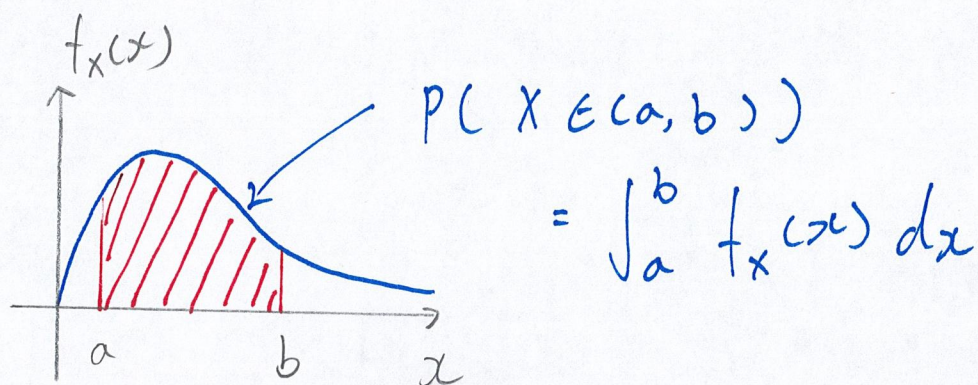


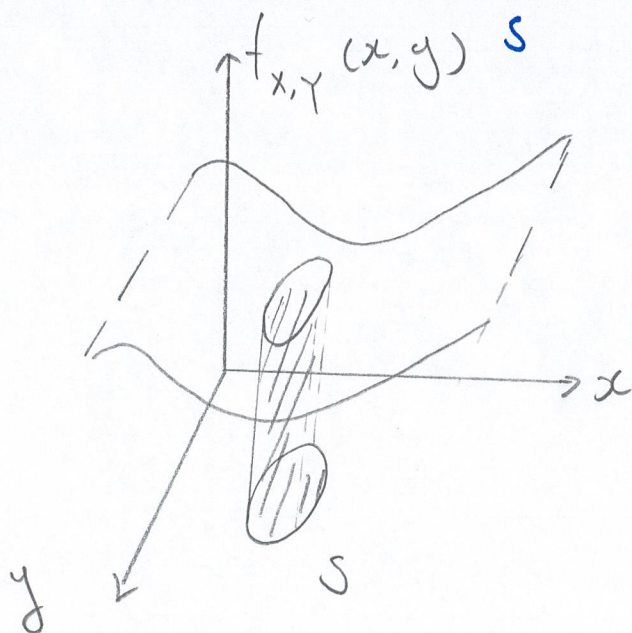
Joint probability density function

▮ In the 1-dimensional case

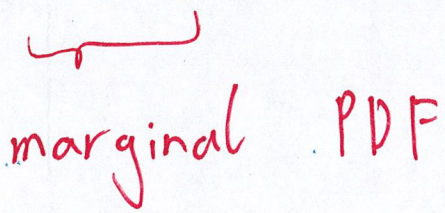


▮ The joint PDF of a continuous random pair (X, Y) is such that

$$P((X, Y) \in S) = \iint_S f_{X,Y}(x, y) dx dy$$



▮ Properties of joint PDF / CDF :

- $F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$
$$= \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x, y) dy dx$$
- $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$

marginal PDF
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx = 1$
- $f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y)$

Ex. Consider

$$f_{X,Y}(x,y) = \begin{cases} k x y^2, & 0 < x < y < 1 \\ 0, & \text{o.w.} \end{cases}$$

Find k such that f is a joint PDF.

Two conditions to check:

$$(1) f_{X,Y}(x,y) \geq 0 \Rightarrow k > 0$$

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$$

$$\Rightarrow \int_0^1 \int_0^y k x y^2 dx dy$$

$$= k \cdot \int_0^1 y^2 \cdot \frac{1}{2} x^2 \Big|_0^y dy$$

$$= k \cdot \int_0^1 \frac{1}{2} y^4 dy = \frac{1}{10} \cdot k = 1$$

$$\Rightarrow \boxed{k = 10}$$

Ex. Let (X, Y) have joint PDF :

$$f_{X,Y}(x,y) = \begin{cases} 1.2(x^2+y) & , 0 < x < 1 \\ & 0 < y < 1 \\ 0 & , \text{ o.w. } \end{cases}$$

① Find $P(X < 0.5, Y \leq 0.4)$

$$F_{X,Y}(0.5, 0.4) = \int_0^{0.5} \int_0^{0.4} 1.2(x^2+y) dy dx$$

$$= 1.2 \int_0^{0.5} x^2 \cdot 0.4 + \frac{1}{2} y^2 \Big|_0^{0.4} dx$$

$$= 1.2 \left(0.4 \cdot \frac{1}{3} x^3 \Big|_0^{0.5} + \frac{1}{2} 0.4^2 \cdot 0.5 \right)$$

$$= 0.063$$

② Find $P(X+Y > 0.8)$.

$P((X, Y) \in S)$ where $S = \{(x, y) \mid x+y > 0.8\}$

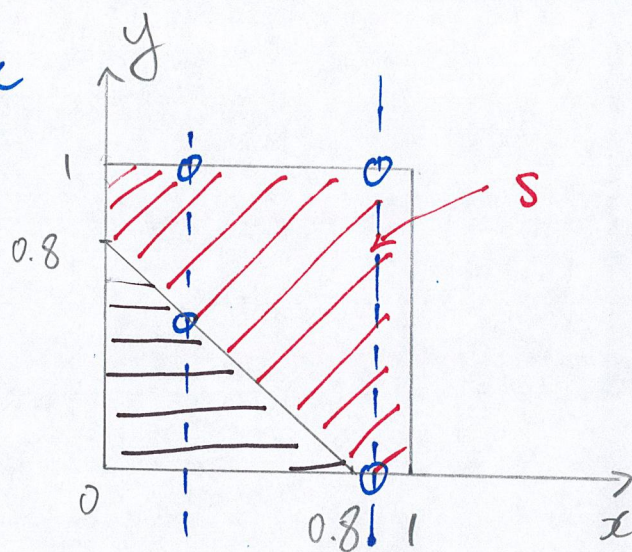
$$= \int_S \int f_{X,Y}(x, y) dy dx$$

$$= \int_0^{0.8} \int_{0.8-x}^1 1.2(x^2+y) dy dx$$

Ⓘ

$$+ \int_{0.8}^1 \int_0^1 1.2(x^2+y) dy dx$$

Ⓙ



$$\text{Ⓘ} = 1.2 \cdot \int_0^{0.8} x^2(0.2+x) + \frac{1}{2}(1-(0.8-x)^2) dx$$

$$= 1.2 \cdot \int_0^{0.8} 0.2x^2 + x^3 + \frac{1}{2} - \frac{1}{2}(0.8-x)^2 dx$$

$$= 1.2 \cdot \left(0.2 \cdot \frac{1}{3} \cdot 0.8^3 + \frac{1}{4} 0.8^4 + \frac{1}{2} \cdot 0.8 - \frac{1}{2} \cdot \frac{1}{3} \cdot 0.8^3 \right)$$

$$\textcircled{I2} = 1.2 \cdot \int_{0.8}^1 x^2 + \frac{1}{2} dx$$

$$= 1.2 \cdot \left(\frac{1}{3} \cdot (1 - 0.8^2) + \frac{1}{2} (1 - 0.8) \right)$$

$$P(X + Y > 0.8) = \textcircled{I1} + \textcircled{I2} = 0.85664$$

Alternatively,

$$P(X + Y > 0.8) = 1 - P(X + Y \leq 0.8)$$

$$= 1 - \int_0^{0.8} \int_0^{0.8-x} 1.2(x^2 + y) dy dx$$

$$= 1 - 0.14366$$

$$= 0.85664.$$